UTSAV KALRA

SYMBIOSIS INSTITUTE OF TECHNOLOGY

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EDUCATION

Qualifications	Institute	Board/University	Year	%/CGPA
B.tech (AIML)	Symbiosis institute of technology	Symbiosis International University	2021-2025	7.5
12th	Ryan international school	CBSE	2020-2021	85%
10th	Ryan international school	CBSE	2018-2019	87%

WORK EXPERIENCE

Associate Software Engineer | upGrad **March 2025-Present**

- Developed an AI-powered AutoGrader system that evaluated diverse assignment submissions (text, images, code, and video) using LlamaIndex, which extracted contextual embeddings through transformer models like BERT.
- Integrated LangChain to interface with GPT-based LLMs, passing rubric-based prompts to dynamically assess student submissions.
- Designed custom prompts for GPT using prompt engineering techniques, enabling it to analyze submissions and generate scores and detailed feedback comments aligned with predefined academic rubrics.
- Automated the complete grading pipeline by combining multimodal embedding, LLMs, and custom evaluation logic, significantly reducing manual evaluation time.
- Gained hands-on experience in transformer architectures, semantic search, LangChain pipelines, and rubric-based evaluation using LLMs.

Tech Stack: Python, BERT, LlamaIndex, LangChain, GPT (OpenAI API), Prompt Engineering, Embedding Models, NLP, Multimodal Data Processing

Data Analyst Intern|Tavant Technologies **July 2024-Sep 2024**

- Leveraged BI tools to analyse and visualize business data, employing preprocessing techniques to clean and transform datasets.
- Conducted exploratory data analysis (EDA) to uncover actionable insights and provided key performance indicators (KPIs) for decision-making.

PROJECTS

Zero-Day Attack Detection Using Autoencoders|SEM VII **Sep-Nov 2024**

- Built an autoencoder-based anomaly detection model integrated with SVM to classify zero-day achieving 75% classification accuracy for zero-day threat detection.
- Optimized model performance by minimizing false positives and enhancing detection precision by 20%

Data Science and Data analysis Project on Human Resource **March-May 2024**

- Data Preparation & Analysis: Cleaned, manipulated, and analyzed HR data using pandas, NumPy, matplotlib, and seaborn to identify key employee trends and patterns.
- Model Development & Evaluation: Built and tested predictive models (Logistic Regression, KNN) using scikit-learn, achieving 98% accuracy in employee retention prediction..
- Business Impact: Delivered actionable insights to enhance HR decision-making and improve employee retention strategies

Insurance Claim Filing Chatbot|SEM V **Oct-Nov 2023**

- Developed a conversational chatbot using DialoGPT and RAG for context-aware insurance claim.
- Integrated Gradio for interactive form submission with structured user input
- Handled 100+ test claims with 95% accuracy in form completion.

SKILLS

Technical skills

- **Programming:** Python, C, C++
- **Databases & BI:** MySQL, Power BI
- **Data Analytics:** EDA, Data Visualization, Statistical Analysis
- **AI & Machine Learning:** Machine Learning, Deep Learning, NLP, Computer Vision, GANs, Feature Engineering
- **Frameworks & Libraries:** NumPy, Pandas, Scikit-learn, PyTorch, TensorFlow, Matplotlib, Seaborn
- **LLM & AI Tools:** BERT, LlamaIndex, LangChain, GPT (OpenAI API), Prompt Engineering

Extra-Curricular Activities

Hackathon Experience

- **Solution Development**: Developed a method to transform traditional X-ray images into MRI/CT-like visuals using Gaussian blur, jitter, and Bilateral filtering to highlight soft tissues, followed by CycleGAN for enhanced image mapping.